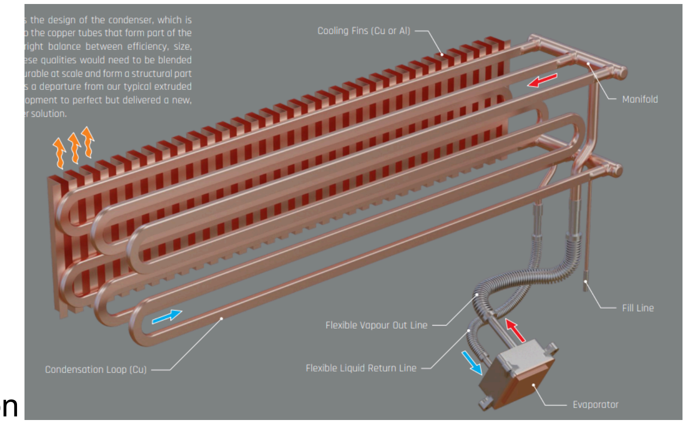
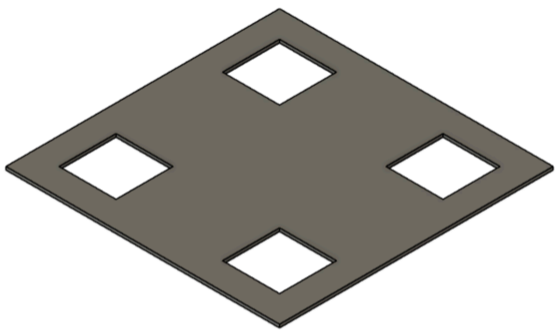
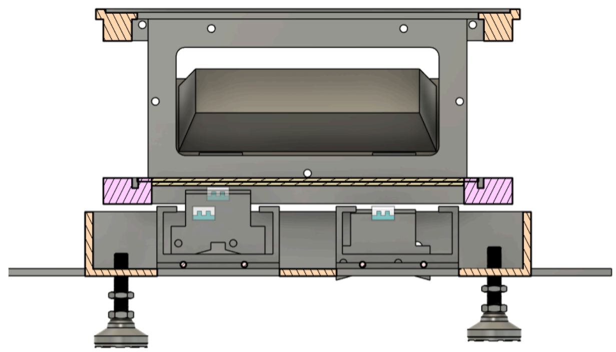
Ethan Kim

- Researched passive cooling technology that could be used for the cool side of our TEG. This one is made for CPUs by Steacom.

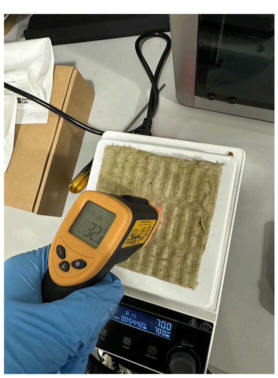


- Made a CAD prototype for putting insulation on the bottom of the copper plate to help keep the cold side section even cooler.

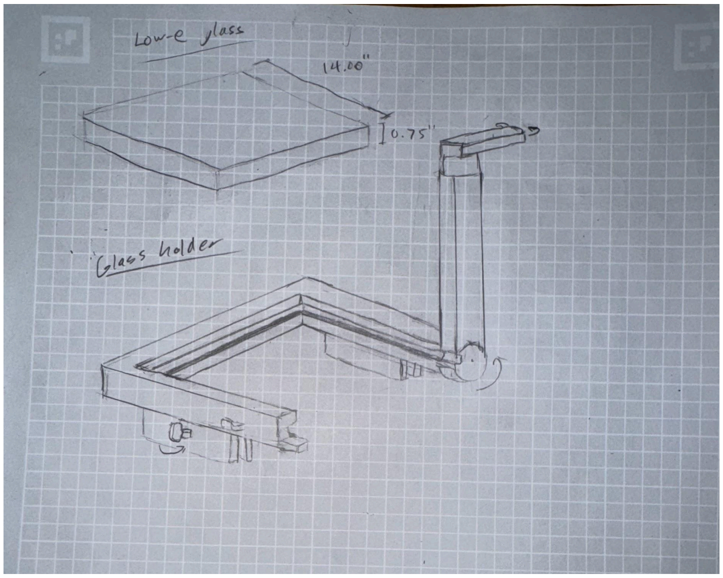




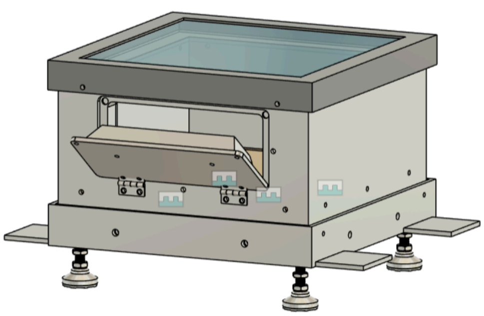
- I researched potential insulation materials and landed on fiberglass and mineral wool insulations. Boyuan and I tested both insulations in the lab. We found a fairly inconclusive difference between the insulation capabilities of each, but mineral wool insulation seems to be better for our purposes due to its sturdier form factor.

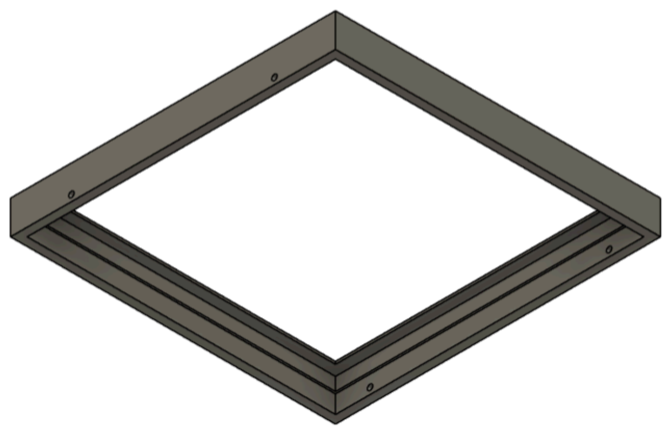


- I sketched a possible glass holder



- Scrapping the previous design for something more simple, I created a CAD prototype and 3D printed a corner of it. It didn’t line up exactly due to differences between the assembly file and the real world oven, but with some modifications it could work well.





- This semester, I learned a lot about how to conduct experiments, prototype efficiently, learned ANSYS, and learned how to research improvements.
- I had a great time during the semester. For the third prototype, I think the insulation could be installed and the glass holder could be fully prototyped.